UniPods METI AI Programme 2026 · Cohort 1

DocTrack

Individual Workplan

November 2026 – October 2028

Digital Lost and Found Document Management in Senegal

Project Lead: Malang Marna

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1. Executive Summary

DocTrack is a platform that helps police stations and gendarmerie units in Senegal to record Loss Declarations and Found Documents, match them, and then return each document to the right person after verification at the Service Counter. Our aim is for the Competent Authority to adopt DocTrack and integrate it into Lost and Found Document Management, supporting the move from manual processes to digital management.

Current situation. The need was established from observations and information gathered from police units; part of this information is confidential and is not reproduced here. DocTrack has a functional foundation that is being consolidated. Several features and integrations still need to be completed and made operational. The support we are seeking will enable us to complete this development, secure its deployment and support the institutional integration of the solution. No police station or gendarmerie unit uses DocTrack yet; the online version is the earlier version (V1).

Overall objective. To consolidate DocTrack, deploy it securely and integrate it progressively into the relevant police stations and gendarmerie units, with the programme's support, with a view to its adoption by the Competent Authority.

Support requested. We are requesting the programme's support in four areas. None of this support has been granted to date.

Area	Content
1. Support in engaging with the authorities (priority)	Introductions, presentations and demonstrations, authorisation process, application for authorised use, data processing framework, collaboration agreement
2. Infrastructure and services over 24 months	Hosting, database, private storage, backups, monitoring, SMS, email, domain name
3. Artificial intelligence (AI) resources	Priority access to application programming interfaces (APIs), tokens, compute and model hosting for Matching, document reading, the voice assistant and understanding user requests
4. Start-up grant and expertise	Professional security audit, targeted support in AI and Wolof, Part-time Support, data protection advice, QR equipment, travel, running costs

Budget over 24 months. The baseline required to get started amounts to **USD 14,440**: USD 1,603 in services requested as In-kind Support and USD 12,838 in monetary grant, of which USD 7,431 is for the security audit and Security Retesting. A conditional extension of **USD 9,378** would be committed only after a positive review. The amounts are based on verified published rates or on estimates that are flagged as such; no quotation has been received yet.

Decisions that rest with the authorities. Authorising the use of DocTrack, designating the police stations and gendarmerie units, setting the data processing framework and deciding on adoption. We are preparing these decisions; we do not control their timing.

2. The Project and its Current Status

2.1 The problem and our response

Today, declared losses and Found Documents are tracked in separate registers, often on paper. A document handed in to a police station or gendarmerie unit may remain there without its owner knowing, while the owner goes through procedures and incurs costs to replace it. The information gathered from police units established this need; it will be used for operational planning with the authorities, with due respect for its confidentiality.

The process we are building:

1. The officer records the Loss Declaration at the Service Counter and issues a certificate of loss.
2. A Found Document is recorded by a police station or gendarmerie unit, or reported by a finder.
3. DocTrack calculates a matching score, explained criterion by criterion; an officer confirms or rejects the match.
4. The owner is notified only after this confirmation.
5. The document is handed over at the Service Counter, after the identity document has been checked, against a verifiable Signed Receipt Acknowledging Return.

Officers and managers at police stations and gendarmerie units are the priority users; citizens are the beneficiaries.

2.2 Status of components

The table distinguishes what works, what is partially developed, and what has been prepared but is not operational. The “working” features have been tested personally in a development environment and through automated tests; these tests do not prove that the platform works under real conditions.

Component	Status at 25 October 2026	What the plan will complete
Works in the development environment		
Loss Declaration at the Service Counter, certificate of loss, recording of Found Documents	Processes tested personally	Production Deployment, audit, use by officers
Rule-based Matching and validation by an officer	Tested on fictitious cases; performance not measured	Measured evaluation, then improvement
Document Return at the Service Counter and verifiable receipt	Tested personally	Procedures validated with the police stations and gendarmerie units

Component	Status at 25 October 2026	What the plan will complete
Accounts, roles and permissions per unit; 221 reference records of police stations and gendarmerie units loaded	Tested personally	Named accounts for authorised officers
Sending emails	The only external integration confirmed as operational	Sender address on our own domain
Partially developed		
Document reading by optical character recognition (OCR)	Local trials on the Project Lead's computer with fictitious documents; no integrated or deployed service	Evaluation, integration, hosting, deployment
Guided assistant in Wolof and French	Dialogue logic written; not tested with users	Voice version v1 integrated and tested
Text-to-Speech and Speech-to-Text	Prototypes; Text-to-Speech output truncated, Speech-to-Text not evaluated	Correction, evaluation, integration
QR labels (two-dimensional barcode)	Creation and revocation in the code; no expiry or scan traceability	Completion, printing, roll-out
Prepared, not operational		
SMS, mobile payment, language models via API	Connectors prepared; no access, credentials or credits; no real-world testing	Access, configuration, real-world testing, validation
V2 going live, Off-site Backups, Monitoring	Prepared; not yet carried out	Secure and monitored deployment

Versions. V1, which is online, is the earlier version of DocTrack: it includes neither the Service Counter process nor the current Matching. V2 brings these features together in our development environment; it has not been deployed. The AI components run as separate services, currently only locally.

2.3 External integrations

The connectors present in the code were prepared in anticipation of future activation. Their presence does not mean that they work. Depending on the case, each one requires access to the service, credentials, credits, completion of development, configuration, real-world testing and validation before use. Changing provider may require adaptations and further testing. The services referred to in the code are not choices imposed on the programme.

2.4 Safeguards for citizens

- **No alert without an officer.** A high score creates a verification task; only human confirmation triggers the alert to the owner.
- **Notifying is not returning.** The alert invites the owner to come in; the handover requires the identity document, a comparison with the photo and, where necessary, a verified power of attorney.
- **Minimal data.** Document numbers are stored as an encrypted fingerprint; the public verification page shows only initials.
- **Vigilance.** Duplicates and fraud signals are reported to a manager, never penalised automatically.

3. Digitisation Objective and Expected Results

3.1 Aim and overall objective

Aim. For the Competent Authority to adopt DocTrack and integrate it into the Lost and Found Document Management system, supporting the move from manual processes to digital management, in accordance with the decisions and conditions it sets. This adoption is not assured.

Overall objective. To consolidate DocTrack, deploy it securely and integrate it progressively into the relevant police stations and gendarmerie units, with the programme's support.

We distinguish two levels of integration: **adoption in procedures** (officers use DocTrack to record, match and return documents) and a possible **technical connection** to existing government systems, which would require specific scoping with the departments concerned.

3.2 Planned progression

1. **Institutional preparation**: presentations, identification of the decision-making process, application for authorised use, data processing framework.
2. **Technical completion**: partial features completed, integrations activated, security audit and fixes.
3. **Secure deployment**: monitored and backed-up Production Deployment.
4. **Authorised use** in one or two police stations or gendarmerie units; running alongside the existing registers may be a transitional step, to be agreed with the authorities.
5. **Support for officers**: training, assistance, visits.
6. **Operational review** shared with the authorities.
7. **Progressive integration** into procedures and extension to other police stations and gendarmerie units, if the authorities so decide.

3.3 Specific objectives

1. Secure reliable infrastructure for 24 months and deploy V2 on it, backed up and monitored.
2. Complete the partial features and activate the necessary integrations (SMS, document reading, voice assistant, QR).

3. Have security audited by a professional, fix the issues and have them retested before any real data is used.
4. Measure and then improve Matching, document reading and the voice assistant on documented cases.
5. Prepare authorised use with the authorities, carry it out, train officers and review the results.
6. Prepare progressive integration and technical and financial continuity beyond 24 months.

3.4 Expected results

The targets are proposed objectives. We distinguish the **meeting** (which we request), the **authorisation** (decided by the authorities), the **installation** (which we carry out once authorised) and **actual use** (officers use DocTrack in their work).

Timeframe	Proposed results	Dependencies
3 months end of January 2027	V2 in production, off-site backup successfully restored, Monitoring active Matching measured on 300 cases; document reading measured on 100 fictitious documents; Text-to-Speech corrected At least two presentation meetings held; decision-making process identified; application for authorised use ready Progress report submitted to the programme	Access to infrastructure; introductions by the programme
6 months end of April 2027	Security audit carried out, critical and high-severity vulnerabilities fixed, Security Retesting completed Voice assistant v1 integrated and tested with 20 volunteers; SMS operational Application for authorised use submitted If authorisation is granted: installation and training in one or two police stations or gendarmerie units	Funding for the audit; SMS credits; decision by the authorities
12 months end of October 2027	If use is authorised: at least 12 weeks of actual use and an operational review (declarations, confirmed matches, document returns, timescales, officer feedback) QR labels in place at volunteer police stations and gendarmerie units Comparison of Matching methods completed Integration and funding proposal presented to the Competent Authority	Authorisation; use by officers

Timeframe	Proposed results	Dependencies
24 months end of October 2028	If the review is positive and the authorities so decide: around 10 police stations and gendarmerie units in one or two regions Collaboration agreement, subject to the authorities' decision Plan for integration into procedures and plan for funded continuity	Positive review; decision by the authorities; sustainable funding

Progress report. At the end of January 2027, we will submit to the programme a report containing a demonstrable version of V2 with fictitious data, the measured results, the status of institutional steps and updated needs, together with a 10-minute presentation. This report will be updated every quarter and will also support our engagement with the authorities.

4. Activities and 24-month Timeline

4.1 Work organisation

Management. Malang Marna, the Project Lead, is responsible for project management, core development and institutional coordination. Work progresses along three parallel tracks — technical, institutional and financial — without waiting for an authorisation or funding where the tracks are independent. Only the professional audit, the purchase of QR equipment and use by officers require a prerequisite.

Prioritised activities. Activities are ranked at three levels. Where trade-offs are needed, the first level takes precedence over the others.

1. **Foundation**: Production Deployment, backups, security and institutional preparation (Activities 01, 02, 04 and 10).
2. **Features and integrations**: completion of partial features, Matching, document reading, voice assistant and SMS (Activities 03, 05, 06, 07 and 08).
3. **Deployment resources**: QR labels, authorised use, funding and progress report (Activities 09, 11, 12 and 13).

Tracking deliverables. A weekly dashboard shows, for each activity, the expected deliverable, its due date, its status and any blockers. A monthly check-in is held with the programme, and the timeline is revised every quarter in the progress report. The workloads shown in the activity sheets are estimates of the Project Lead's working time.

4.2 Progressive mobilisation of contributors

Three complementary contributions will support the Project Lead when they are needed. These are resources still to be mobilised: no contributor has been recruited to date. They are distinct from the mentoring requested from the programme, which is advisory and is not budgeted.

Contribution 1 · Specialised security services	
Tasks	Architecture and configuration audit; authorised Penetration Testing within a written scope; prioritised report; Security Retesting after our fixes; targeted test after major changes
Period	February to April 2027; targeted test during the extension, in 2028
Deliverables	Audit report, retesting report, then targeted test report
Estimated workload	Set by the provider's scope and quotation; around 40 hours of fixes for the Project Lead
Funding	Existing lines: audit (USD 6,406) and Security Retesting (USD 1,025) in the baseline; targeted test (USD 1,708) in the extension · Quotation Required

Contribution 2 · Targeted support in AI, voice integration and Wolof evaluation	
Tasks	Integration of Speech-to-Text and Text-to-Speech models; running model-based comparisons for Matching and document reading; reference transcription of Wolof recordings; review of the assistant's phrases; scoring of tests with volunteers
Period	December 2026 to April 2027, then during voice assistant extensions from November 2027
Deliverables	Annotated reference corpus, validated Wolof phrases, evaluation report on voice and model-based comparisons
Estimated workload	Around 100 hours in the first period: 60 hours for Wolof evaluation, 40 hours for integration and comparisons
Funding	Not yet costed: no existing budget line covers this work; amount to be established on the basis of a quotation (section 8.6). Compute resources and tokens remain in category A

Contribution 3 · Part-time Support for deployment, training and officer support	
Tasks	Help with installation and account creation; running training sessions with the Project Lead; assistance to officers; keeping the incident log; gathering feedback for the review
Period	Six months from the start of authorised use; twelve months during the extension
Deliverables	Updated training materials, support log, monthly reports
Estimated workload	Part-time, around eight days per month (estimate)
Funding	Existing Part-time Support lines: USD 769 in the baseline and USD 2,050 in the extension · estimate

4.3 November 2026 to January 2027, by fortnight

Period	Priorities	Prerequisite
2 to 15 November	Production Deployment of V2 (Activity 01); off-site backup and first restore test (Activity 02); request for introductions to the programme (Activity 10); correction of Text-to-Speech (Activity 07)	Programme infrastructure or current hosting
16 to 29 November	Monitoring; security hardening (Activity 04); set of 300 cases and baseline measurement (Activity 05); presentation pack; voice recordings from consenting volunteers	None
30 November to 13 December	Presentations and demonstrations; decision-making process; Speech-to-Text test bench; evaluation of document reading (Activity 06); grant application (Activity 12); requests for security quotations and search for targeted Wolof support	Introductions obtained
14 to 27 December	Data processing framework; adjusted Matching variant; voice assistant prototype; completion of QR expiry and scan history (Activity 09)	Data protection advice
28 December to 10 January	Reduced period: internal voice tests; first real SMS (Activity 08)	SMS credits
11 to 24 January	Application for authorised use finalised; voice assistant tests with 10 volunteers; officer training materials	None
25 to 31 January	Progress report (Activity 13); three-month review	None

4.4 Activity sheets

Activity 01 · Production Deployment and stabilisation of V2	
Tasks	Rehearsal on a copy of the V1 data; migration; scheduled tasks; putting the application programming interface and then the website online; fixes during the first four weeks
Period and workload	November and December 2026 · 25 hours
Support requested	Hosting (area 2); otherwise, current hosting without document reading
Dependencies	None to start
Deliverable	V2 online; operations guide
Indicator and success criterion	Monthly availability · no data loss during migration

Activity 02 · Backups, restoration and Monitoring	
Tasks	Daily encrypted copy stored off the server; full restoration on a clean machine; availability and error alerts
Period and workload	November 2026, then a test every quarter · 10 hours
Support requested	Private storage and Monitoring (area 2)
Dependencies	Activity 01
Deliverable	Written and tested restoration procedure
Indicator and success criterion	Restoration time · successful in under 2 hours

Activity 03 · Completion of partial features	
Tasks	Complete the processes that are still incomplete; fix the issues found during personal testing; document the features for officers
Period and workload	December 2026 to March 2027 · 40 hours
Support requested	None to start
Dependencies	Activity 01
Deliverable	Consolidated version and release notes
Indicator and success criterion	Open issues · no blocking issue before authorised use

Activity 04 · Security: hardening, professional audit, fixes	
Tasks	Hardening in line with recognised web application security standards; audit by a specialised provider: architecture, configuration, authorised Penetration Testing within a written scope, prioritised report; fixes; Security Retesting by the provider
Period and workload	Hardening in November 2026 (15 hours); audit and fixes from February to April 2027 (an estimated 40 hours of fixes)
Support requested	Grant for the audit (area 4); Contribution 1
Dependencies	Activity 01; funding
Deliverable	Audit report, remediation plan, retesting report
Indicator and success criterion	Vulnerabilities by severity · no open critical or high-severity vulnerability before real data is used

Activity 05 · Evaluation and improvement of Matching	
Tasks	Set of 300 labelled fictitious cases; baseline measurement; adjusted variant; comparison with a trained model and with language-model assistance on the same cases (section 5.2)
Period and workload	November 2026 to January 2027 (35 hours); full comparison from February to July 2027
Support requested	Compute and tokens (area 3); Contribution 2 for model-based comparisons; programme mentoring to review the protocol
Dependencies	None for the baseline measurement
Deliverable	Reproducible evaluation report
Indicator and success criterion	False matches, missed matches, cost, response time · no false match above the alert threshold

Activity 06 · Document reading (OCR): evaluation, integration, deployment	
Tasks	Test bench of 100 fictitious documents (skewed photos, glare, identity cards, passports); field-by-field measurement; integration into the officer's workflow; hosting of the service; comparison with an external service on fictitious data
Period and workload	December 2026 to April 2027 · 30 hours
Support requested	Hosting for AI components, OCR credits (area 3)
Dependencies	Infrastructure
Deliverable	Evaluation report; integrated and deployed service
Indicator and success criterion	Share of correct fields, data entry time · made available to officers if it reduces data entry without adding errors

Activity 07 · Voice assistant in Wolof and French	
Tasks	Correction of Text-to-Speech; Speech-to-Text test bench; understanding of requests; confirmation; integration into the citizen journey; tests with volunteers (section 5.4)
Period and workload	November 2026 to April 2027 · 75 hours
Support requested	GPU compute, Speech-to-Text, tokens (area 3); Contribution 2; volunteer speakers
Dependencies	Recordings made with consent
Deliverable	Voice assistant v1 online; test report
Indicator and success criterion	Task success rate, transcription errors · targets set after the first test bench

Activity 08 · Activation of SMS and notifications	
Tasks	Access to the service, credentials and credits; configuration; real-world testing; delivery tracking; email sender address on our own domain
Period and workload	December 2026 and January 2027 · 8 hours
Support requested	SMS credits, domain name (area 2)
Dependencies	Activity 01
Deliverable	Operational SMS; delivery tracking table
Indicator and success criterion	Delivery rate · more than 95% of SMS delivered during tests

Activity 09 · Roll-out of QR labels	
Tasks	Complete revocation, renewal, expiry and scan traceability; printing from the Service Counter; checks on readability, adhesion and durability; validation of the labelling procedure with the police stations and gendarmerie units (section 6)
Period and workload	Development from December 2026 to February 2027 (20 hours); roll-out during authorised use
Support requested	Printers and consumables (area 4)
Dependencies	Funding; agreement of the police stations and gendarmerie units
Deliverable	Complete QR feature; printing and labelling procedure
Indicator and success criterion	Readability after three months, QR codes scanned, incidents · no observed damage to documents

Activity 10 · Institutional preparation and authorised use	
Tasks	Request introductions; present and demonstrate; identify the decision-making process; draft the application for authorised use (purpose, units, data, security, duration, exit arrangements); submit it; prepare the data processing framework and a draft agreement
Period and workload	November 2026 to January 2027 (30 hours), then ongoing
Support requested	Introductions and support from the programme (area 1)
Dependencies	Availability of contacts; decisions by the authorities
Deliverable	Application for authorised use; meeting reports; draft agreement
Indicator and success criterion	Meetings held, application submitted · written decision by the authorities (timing outside our control)

Activity 11 · Authorised use, support for officers and review	
Tasks	Installation; named accounts; two-hour training session; one-page guide; assistance; weekly visit in the first month; incident log; review shared with the authorities
Period and workload	From authorisation, for at least 12 weeks · 6 hours per week
Support requested	Travel, connectivity (area 4); Contribution 3
Dependencies	Authorisation; Activity 04 completed
Deliverable	Operational review
Indicator and success criterion	Indicators in section 9.1 · no document returned to the wrong person

Activity 12 · Start-up grant and funding model	
Tasks	Grant application; search for suitable funding; full cost per police station or gendarmerie unit; identification of the paying body and its procurement procedure
Period and workload	November 2026 to January 2027 (10 hours), then throughout 2027
Support requested	Guidance from the programme; mentoring on public procurement (area 4)
Dependencies	None to start
Deliverable	Grant application; funding model note
Indicator and success criterion	Applications submitted · funding for the audit obtained before the end of March 2027

Activity 13 · Progress report	
Tasks	Online demonstration, measured results, status of institutional steps, updated needs, 10-minute presentation
Period and workload	End of January 2027 (12 hours), then updated every quarter
Support requested	None
Dependencies	Activities 01 to 10
Deliverable	Report and presentation
Indicator and success criterion	Report submitted on 31 January 2027

4.5 February 2027 to October 2028, by quarter

Period	Priorities	Condition
February to April 2027	Mobilisation of the security provider (Contribution 1): audit, fixes and Security Retesting; targeted Wolof support (Contribution 2); voice assistant v1; document reading integrated; installation and training if use is authorised, with Part-time Support (Contribution 3)	Funding; authorisation
May to July 2027	Actual use and weekly follow-up; roll-out of QR labels; comparison of Matching methods	Authorised use under way
August to October 2027	Operational review; integration and funding proposal; draft collaboration agreement	12 weeks of use
November 2027 to January 2028	Extension to three to five police stations or gendarmerie units with Part-time Support; new tasks for the voice assistant	Positive review; decision by the authorities
February to April 2028	Continued extension; trained Matching model if there are enough officer decisions	Funding for the extension
May to July 2028	Around 10 police stations and gendarmerie units; targeted security test (Contribution 1)	As above
August to October 2028	Plan for integration into procedures; continuity plan; 24-month review	None

5. AI Development and Accessibility

5.1 Priority AI resources

AI serves four priority functions. Each is measured before being made available, and none decides on its own whether a document is returned. We are requesting priority access to the following resources:

Function	Role for the user	Resources needed
Matching	Find the owner despite spelling errors, name variants and namesakes	Compute and tokens for comparisons; storage of case sets
Document reading and structuring	Record a Found Document from a photo	CPU model hosting; reading API for comparison
Speech-to-Text and Text-to-Speech	Serve citizens who are less comfortable with reading and writing	GPU compute; Speech-to-Text API; model hosting; audio storage
Understanding and assistance	Understand a free-form request, guide, confirm	Language model tokens; model hosting

5.2 Matching: comparing methods on the same cases

The current Rule-based Matching is our **baseline**. Improvement through a model is an objective to be developed and measured: no better performance has been demonstrated to date.

Method	Principle	Condition
Current rules	Name similarity, phonetic key, Senegalese name spellings, date of birth, document number, dates, region	Measured first
Adjusted rules	Weights and thresholds revised in light of observed errors	After the baseline measurement
Trained model	Ranking learned from the sub-scores and officers' decisions	Enough validated decisions
Language-model assistance	Opinion on ambiguous cases only	Fictitious data, or a model hosted under our control

Protocol. 300 labelled fictitious pairs in six groups: spelling variants, namesakes, siblings, missing fields, data entry or reading errors, swapped first and last names. For each method and each threshold, we measure false matches (the most serious), missed matches, the cost per 1,000 comparisons and the response time. Targets will be set after the first measurement. Validation by an officer remains mandatory, and no real data is sent to an external service.

5.3 Document reading (OCR)

Document reading suggests the fields; the officer checks them before saving. Images sent for reading are deleted after one hour. The local trials do not constitute an operational service: evaluation, integration into the officer's workflow and deployment on the infrastructure are needed before the service is made available.

5.4 Voice assistant in Wolof and French

The voice assistant is part of our value proposition: it opens DocTrack to citizens who find it difficult to read or write French. It guides the user; it does not replace the officer.

Step	Content	Period
Prototype correction	Fix the truncated Text-to-Speech output; regenerate the fixed phrases in Wolof and French	November 2026
Speech-to-Text	Test bench in Wolof and French on recordings made with consent, using invented data; choice of model	November and December 2026
Understanding	Existing rules and keywords; language model for free-form sentences, with numbers masked	December 2026 to February 2027

Step	Content	Period
Confirmation	The assistant repeats what it has understood and asks “yes” or “no” before acting	January 2027
Voice response	Pre-recorded fixed phrases; no personal data sent for speech synthesis	January 2027
Integration and testing	Citizen journey; tests with 10 and then 20 volunteers: task success, understanding	January to April 2027
Extensions	Other tasks, other dialects, telephone or messaging channel	From November 2027

Initial scope. Find the nearest police station or gendarmerie unit and know what to bring; prepare a preliminary Loss Declaration, which is then validated at the Service Counter; report a Found Document. **Fallback.** After two failed attempts to understand, the assistant offers the written form or the unit’s contact details. **Identity.** Voice is never enough to establish identity or open a personal file: access to a file is through the citizen’s account, and handover is at the Service Counter.

6. QR Labels and Operational Resources

6.1 Function and controls

The QR label is a measure for prevention and for making it easier to report a Found Document; it is separate from the loss and Document Return process. A citizen links a QR code to one of their documents; a finder who scans it indicates where it was found; the owner is notified and the nearest police station or gendarmerie unit can arrange collection.

Already present in the code: one QR code per document, a scan page that displays no personal data about the owner, and a first version of revocation and renewal.

We will put in place:

- completed and tested revocation and renewal, with the old QR code deactivated;
- an expiry date for each QR code;
- traceability of scans: date, unit or approximate area, without keeping more than necessary;
- a history that can be viewed only by the owner and, for collection purposes, by authorised officers of the police station or gendarmerie unit concerned.

6.2 Placement on the document

The label is a very small square placed directly on an available area of the identity document or other document, without covering the information, photo, signatures or any reading and security features. A protective sleeve remains optional. The exact size will be set according to the QR code’s actual readability. We will check printing, adhesion and the absence of damage. The labelling procedure will be submitted for

validation by the police stations, gendarmerie units and authorities concerned; it has not yet been authorised.

6.3 Roll-out of QR labels

Equipment is provided progressively: one printer for development and quality checks, then one printer for each police station or gendarmerie unit that offers the QR label, when it starts using it. Consumables follow the volumes observed (three rolls of 1,000 labels at the start). Maintenance includes cleaning print heads and replacing worn parts. Readability and durability checks are ongoing quality tasks, not an optional trial.

6.4 Workspace and travel

We are not requesting dedicated premises at the start. The UniPod in Diamniadio, a university innovation hub supported by the United Nations Development Programme (UNDP), could offer occasional access to a workspace, equipment or support, subject to availability. As the Project Lead is based in Dakar, each trip to Diamniadio has a cost and takes time. We will therefore work remotely wherever possible, group together sessions that require attendance in person, and plan occasional trips for meetings, demonstrations and use of equipment. This travel has its own budget line, separate from site visits to police stations and gendarmerie units.

We do not assume that every police station or gendarmerie unit is equipped: we are providing 4G modems for a unit without a connection and an entry-level Android device to test DocTrack under users' conditions.

7. Support Requested

We describe capabilities, not providers: the programme can negotiate the most suitable solutions. None of this support has been granted to date; the 24-month services are a negotiation objective for the programme.

7.1 Support in engaging with the authorities

What we request from the programme	What we deliver	What rests with the authorities
Introductions to decision-makers (police, gendarmerie, ministry, the national Personal Data Protection Commission)	30-minute presentation and demonstration	Agree to a meeting
Help identifying the authorisation process	Application for authorised use	Authorise, designate the police stations and gendarmerie units
Review of the application and support with its submission	Responses to requests for further information	Set the conditions and duration
Help clarifying responsibilities and data processing	Proposed roles, retention periods and security measures	Designate the data controller and complete the formalities
Scoping of a collaboration agreement	Draft agreement	Negotiate and sign

What we request from the programme	What we deliver	What rests with the authorities
Preparation for progressive integration	Training, support, review	Decide on adoption and extension

7.2 Infrastructure and technical services

Priority: high (H) or medium (M). We will keep a monthly consumption table shared with the programme.

Capability	Use and estimated volume	Prio.	After 24 months
Application hosting	Application, database, scheduled tasks; 2 then 10 police stations or gendarmerie units	H	Included in the service cost
Hosting for AI components	Document reading, CPU-based Speech-to-Text	H	As above
Private storage	Photos, documents, audio; 100 GB then 500 GB	H	Deletion in line with retention periods
Backup and restoration	Daily encrypted off-site copy; quarterly test	H	Marginal cost
Monitoring	Availability, errors, logs	H	Marginal cost
SMS	Alerts to citizens without email; 20,500 then 120,000	H	Included in the service cost
Transactional email	Activation, alerts, passwords	M	Low cost
Domain name and certificates	National web address; automated certificates	M	Annual renewal

7.3 AI resources

Resource	Use and estimated volume	Prio.	After 24 months
Language model tokens	Understanding, assistance, trials; 1,000 conversations per month	H	Or a hosted open model
Speech-to-Text via API	Wolof and French; 1,000 minutes per month	H	Depending on the test bench
GPU compute	Speech synthesis, evaluation and adaptation of voice models; 150 then 450 hours	H	Models served on CPU where possible

Resource	Use and estimated volume	Prio.	After 24 months
Document reading API	Comparison and fallback; 6,000 then 36,000 pages	M	Optional

7.4 Grant and specialised expertise

We are requesting a Start-up and Operating Grant, or guidance towards suitable funding, for items that In-kind Support does not cover: equipment and set-up (QR printers, test device, modems); recurring running costs (connectivity, travel, consumables, helpline); cash flow before the first revenues (Part-time Support, Contribution 3); specialised services (Contribution 1). The targeted support in AI and Wolof (Contribution 2) has yet to be costed.

Security audit. This is a professional engagement, not an informal review: architecture and configuration audit, authorised Penetration Testing within a written scope, a report prioritised by severity, fixes, then Security Retesting by the provider. This item accounts for more than half of the baseline grant, because no real citizen data will be processed without it.

Mentoring. Applied AI (review of evaluation protocols, document reading, voice); security; public procurement and funding model; data protection. This advice does not replace the paid contributions described in section 4.2 and is not budgeted.

8. Budget in USD

8.1 Method

All amounts are in US dollars (USD). Rates published in CFA francs were converted at the BCEAO (Central Bank of West African States) reference rate of 22 October 2026 (USD 1 = 585.41 CFA francs); rates published in euros were converted using the BCEAO's fixed parity applied to the same rate. Totals are calculated on exact values and then rounded to the nearest dollar; a sum of rounded lines may therefore differ by one dollar.

Each line shows its status: **verified rate** (published rate checked on 23 October 2026), **estimate** (cannot be verified at this stage) or **quotation required**. No quotation has been received. The baseline covers the needs required to get started; the extension is committed only after a positive review. Both cover November 2026 to October 2028.

8.2 Summary

Category	Baseline	Extension
A. Resources and services requested as In-kind Support (estimated value)	USD 1,603	USD 2,605
B. Monetary grant requested	USD 12,838	USD 6,774
Total requested (A + B)	**USD 14,440**	**USD 9,378**
C. Confirmed resources already available	Not valued	—

Category	Baseline	Extension
D. Additional needs to be funded	Section 8.6	—

The targeted support in AI and Wolof (Contribution 2) has not yet been costed and is not included in these totals.

8.3 A. Resources and services requested as In-kind Support

The value shown is that of the service at its published reference rate. If the programme secures these services, they will not be purchased or funded a second time.

Baseline

Item	Quantity and duration	Basis of calculation	Amount	Status
Hosting for the application and database	1 server (2 virtual CPUs, 8 GB memory, 100 GB), 24 months	USD 9.99/month in year 1, USD 16.99/month in year 2	USD 324	Verified rate
CPU hosting for AI components (OCR, speech-to-text)	1 server of the same capacity, 24 months	Same reference rate	USD 324	Verified rate
Private storage and off-site backups	100 GB, 24 months	USD 6.95 per TB per month	USD 17	Verified rate
Monitoring (availability, errors, logs)	1 service, 24 months	USD 5/month	USD 120	Estimate
Transactional email	1 subscription, 24 months	USD 9/month (reference paid tier)	USD 216	Verified rate
SMS credits	20,500 SMS	Operator bundles of 20,000 and 500 SMS	USD 149	Verified rate
National domain name and certificates	2 years	USD 42.71/year (top of the published range); free automated certificates	USD 85	Estimate
Graphics processing unit (GPU) compute for voice	150 hours	USD 0.80/hour	USD 120	Verified rate
Language model tokens	209 M input and 30.2 M output	USD 0.25 and USD 2.00 per million	USD 113	Verified rate
Speech-to-text via API	21,000 minutes	USD 0.006/minute	USD 126	Verified rate
OCR via API (comparison and fallback)	6,000 pages	USD 1.50 per 1,000 pages	USD 9	Verified rate
Total			**USD 1,603**	

AI assumptions: one conversation consists of 6 exchanges of 1,500 input tokens and 200 output tokens; 1,000 conversations per month from February 2027 to October 2028, plus 20 million input tokens and 5 million output tokens for trials; one minute of audio per conversation. Sources: Hostinger KVM 2 listing (VPSBenchmarks), Backblaze B2, Brevo, Orange Senegal SMS bundles, GPT-5 mini, Whisper, Google Cloud Vision, Hugging Face Inference Endpoints, NIC Senegal (Kolonell), checked on 23 October 2026.

Conditional extension

Item	Quantity and duration	Basis of calculation	Amount	Status
Second server (testing or redundancy)	1 server, 24 months	Same reference rate	USD 324	Verified rate
Additional GPU compute (live speech-to-text, model adaptation)	450 hours	USD 0.80/hour	USD 360	Verified rate
Additional tokens	972 M input and 129.6 M output, 12 months	USD 0.25 and USD 2.00 per million	USD 502	Verified rate
Additional speech-to-text	108,000 minutes, 12 months	USD 0.006/minute	USD 648	Verified rate
Additional OCR via API	36,000 pages, 12 months	USD 1.50 per 1,000 pages	USD 54	Verified rate
Additional storage	400 GB, 12 months	USD 6.95 per TB per month	USD 33	Verified rate
Additional SMS credits	120,000 SMS, 12 months	Two operator bundles of 60,000 SMS	USD 683	Verified rate
Total			**USD 2,605**	

Assumption: around 10 police stations and gendarmerie units, i.e. 9,000 additional conversations and 9,000 additional minutes of audio per month for 12 months; 10 units × 300 cases × 3 SMS × 12 months = 108,000 SMS.

8.4 B. Monetary grant requested

Baseline

Item	Quantity and duration	Basis of calculation	Amount	Status
Professional security audit	1 engagement	Midpoint of the range published in Dakar for a multi-role application	USD 6,406	Quotation required
Security retesting after fixes	1 engagement	Midpoint of the published range	USD 1,025	Quotation required
Data protection advice	1 flat fee	Budget envelope	USD 683	Estimate

Item	Quantity and duration	Basis of calculation	Amount	Status
QR label printers	2 (set-up and first site)	USD 280.13 each, observed price excluding shipping and taxes	USD 560	Estimate
Rolls of small square labels	3 rolls of 1,000	USD 14.51 per roll	USD 44	Verified rate
Printer maintenance	24 months	USD 50 per year	USD 100	Estimate
Site visits (Dakar)	60 trips	USD 8.54 per trip	USD 512	Estimate
Travel to the UniPod in Diamniadio	24 return trips	USD 8.54 (regional train and final leg)	USD 205	Estimate
Connectivity for the Project Lead	24 months	USD 17.08/month	USD 410	Estimate
4G modems and data plans for sites without a connection	2 modems, 12 monthly plans	USD 42.71 per modem, USD 17.08 per plan	USD 290	Estimate
Entry-level Android device (testing)	1	Budget envelope	USD 171	Estimate
Officer training (materials, sessions)	2 sessions	USD 42.71 per session	USD 85	Estimate
Part-time support: deployment, training, officer support	6 months	USD 128.12/month	USD 769	Estimate
Running costs (helpline, bank charges, supplies)	24 months	USD 17.08/month	USD 410	Estimate
Contingency	10% of the grant	Price variations, shipping, taxes	USD 1,167	Calculation rule
Total			**USD 12,838**	

Sources: ranges published by a security services provider in Dakar (Kolonell, 2026) for the audit and Security Retesting; prices observed in October 2026 for label printers (Hellopro, USD 224 to USD 392) and rolls (PrintAbout); regional train fare between Dakar and Diamniadio of USD 0.85 to USD 2.56 depending on zones (SenCampus, 2026). The other amounts are estimates.

Conditional extension

Item	Quantity and duration	Basis of calculation	Amount	Status
Additional QR printers	5, according to sites added	USD 280.13 each	USD 1,401	Estimate

Item	Quantity and duration	Basis of calculation	Amount	Status
Label rolls	10 rolls of 1,000	USD 14.51 per roll	USD 145	Verified rate
Targeted security test after major changes	1 engagement	Budget envelope	USD 1,708	Quotation required
Field missions outside Dakar	4 missions	USD 128.12 per mission	USD 512	Estimate
Additional officer training sessions	8 sessions	USD 42.71 per session	USD 342	Estimate
Part-time support: deployment, training, officer support	12 months	USD 170.82/month	USD 2,050	Estimate
Contingency	10% of the grant	Price variations, shipping, taxes	USD 616	Calculation rule
Total			**USD 6,774**	

8.5 C. Confirmed resources already available

- The Project Lead's development and management time, computer and development environment.
- The V2 code, its documentation, its automated tests and the Production Deployment guide.
- The current V1 hosting (already paid for) and free hosting for the website: a fallback option, without document reading.
- Sending emails through an existing mailbox, within a limit of 500 messages per 24 hours.
- The programme's training and support.

8.6 D. Additional needs to be funded

- **Not yet costed:** targeted support in AI, voice integration and Wolof evaluation (Contribution 2, around 100 hours in the first period), to be established on the basis of a quotation before being added to the request.
- **Outside this request:** the Project Lead's remuneration; premises; security fixes contracted to a provider if they exceed our skills; annual audits after 24 months.
- **To be mobilised if the programme cannot cover them:** any lines in categories A and B that are not granted. The Project Lead cannot fund them all alone; we will approach other sources (grants, technical partners, a future paying body).
- **Possible adjustments, in this order:** a single QR printer at the start; Part-time Support reduced to three months; tokens and Speech-to-Text limited to trials; fallback hosting without document reading. The security audit and backups will not be reduced before real data is used.

9. Tracking, Risks and Continuity

9.1 Indicators

We will update a dashboard every week, hold a monthly check-in with the programme and submit a report at 3, 6, 12 and 24 months. Usage figures will come from the platform, which already records every status change and every message sent.

Indicator	Source	Proposed target
Service availability	Monitoring	Measured from Production Deployment; target set after one month
Backup restoration	Quarterly test	Successful in under 2 hours
Open critical or high-severity vulnerabilities	Audit and Security Retesting	None before real data is used
False matches above the alert threshold	Set of 300 cases	None
Missed matches	Set of 300 cases	Set after the baseline measurement
Correctly read fields	Test bench of 100 fictitious documents	Set after the first measurement
Tasks completed by voice	Tests with volunteers	Set after the first test bench
Meetings, application, decision	Meeting reports	Two meetings and application ready at 3 months
Officers trained, weeks of actual use	Platform, attendance sheets	12 weeks per police station or gendarmerie unit
Documents returned thanks to DocTrack	Platform	At least one during authorised use
Documents returned to the wrong person	Incident log	None
Time taken to record a declaration	Timing	No longer than with the register
Consumption of services	Monthly table	Within the negotiated allocation
Monthly cost per case	Invoices and volumes	Calculated at 6 and 12 months

9.2 Risks

Risk	Likelihood	Impact	Mitigation
Authorisation delayed or refused	High	High	Application prepared early; programme support; technical work continues
24-month services not obtained	Medium	Medium	Fallback hosting; adjustments in section 8.6
Audit not funded	Medium	High	No real data without an audit; other funding sought

Risk	Likelihood	Impact	Mitigation
Document returned to the wrong person	Low	Very high	Validation by an officer, check at the Service Counter, immediate halt in the event of an incident
Data breach	Low to medium	Very high	Minimal data, encrypted document numbers, audit, encrypted backups
Integration takes longer than planned	High	Medium	Step-by-step activation; real-world testing before each release
Voice assistant misunderstands	Medium	Medium	Systematic confirmation, written fallback, no access to a file by voice
QR label damages a document	Low	High	Adhesion and durability checks; procedure validated with the police stations and gendarmerie units
Workload concentrated on the Project Lead	High	Medium	Three-level prioritisation, progressive mobilisation of the three contributions, weekly tracking of deliverables
Price increases	High	Low	Renewal prices included in the 24-month reference rates
No paying body	Medium	High	Several options explored; no extension without a written commitment

9.3 Continuity after 24 months

The continuity we are aiming for is the integration of DocTrack into the Competent Authority's procedures, if it so decides. The recurring technical cost remains modest in relation to the service provided (hosting, backups, Monitoring, SMS, domain name). It could be covered by a service contract, funding for the extension or the transfer of hosting to public infrastructure. Any technical connection to government systems would require specific scoping. The code is documented so that another team can take it over if necessary.

10. Glossary

Term	Definition
API	Application programming interface: the means by which two software systems exchange data or services.
Tokens	Units of text processed by a language model; billing is per million tokens.
Language model	AI program capable of understanding and producing text.
OCR	Optical character recognition: automatic reading of the text in an image.

Term	Definition
QR code	Two-dimensional barcode that can be read by a phone.
GPU	Graphics processing unit, suited to the calculations of voice models.
Hosting	Server, located in a data centre, on which the application runs.
Backup and restoration	Regular copying of data, and the ability to bring it back into service after an incident.
Monitoring	Automatic surveillance of availability and errors, with alerts.
Security audit	Professional review of the architecture, configuration and code to identify vulnerabilities.
Penetration Testing	Simulated, authorised attacks within a written scope, to test the system’s resilience.
Matching	Linking a Loss Declaration to a Found Document.
Digitisation of Paper-based Processes	Moving from paper-based procedures to digital procedures.
V1 and V2	Earlier version of DocTrack, online; new version, being consolidated.